OXFORD

INTERNATIONAL
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INTERNATIONAL A-LEVEL
MATHEMATICS

(9660/MA04) Unit S2 Statistics

Tuesday 21 January 2020 07:00 GMT Time allowed: 1 hour 30 minutes

Materials
- For this paper you must have the Oxford International AQA booklet of formulae and statistical tables (enclosed).
- You may use a graphics calculator.

Instructions
- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.

Advice
- Unless stated otherwise, you may quote formulae, without proof, from the booklet.
- Show all necessary working; otherwise marks may be lost.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
TOTAL	

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Answer **all** questions in the spaces provided.

1 The continuous random variable X_1 has mean 2 and variance 3

The continuous random variable X_2 has mean 5 and variance 1

The continuous random variable X_3 has mean 1 and variance 0.5

X_1, X_2 and X_3 are independent.

1 (a) Find $\mathrm{E}\left(\sum_{i=1}^{3} X_i\right)$

[1 mark]

Answer _____

1 (b) Find $\mathrm{Var}\left(\sum_{i=1}^{3} X_i\right)$

[1 mark]

Answer _____

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1 (c) Find $\mathrm{E}\left(3X_1 - 4X_3\right)$

[2 marks]

Answer

1 (d) Find $\mathrm{Var}\left(5X_1 - 2X_2\right)$

[2 marks]

Answer

6

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2 The random variable X has an exponential distribution with parameter λ, where $\lambda > 0$

The variance of X is 400

2 (a) Find λ.

[2 marks]

Answer

2 (b) Find the mean of X.

[1 mark]

Answer

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2 (c) Find $\mathrm{P}(X > 25)$, giving your answer to three significant figures.

[3 marks]

Answer

2 (d) Find x such that $\mathrm{P}(X < x) = 0.6$, giving your answer to three significant figures.

[2 marks]

Answer

8

Turn over ►

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3 Matthew is shooting arrows at a target.

The number of arrows hitting the target can be modelled by a binomial distribution with

$n = 100$ and $p = 0.01$

3 (a) Using a Poisson distribution as an approximation, estimate the probability that more than 3 arrows hit the target, giving your answer to three decimal places.

[3 marks]

Answer _____

3 (b) State under what conditions the Poisson distribution is considered to be a good approximation to the binomial distribution.

[2 marks]

5

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4 Let X represent the number of people arriving at a hospital with a particular disease in a day.

A random sample of 20 days is taken. The summarised data is

$$\sum x = 50 \qquad \text{and} \qquad \sum x^2 = 173$$

4 (a) Using the summarised data, explain why it would be reasonable to model X using a Poisson distribution.

[4 marks]

4 (b) Using a Poisson model with mean 2.5, find $\mathrm{P}(X = 4)$, giving your answer to three significant figures.

[2 marks]

Answer _____

6

Turn over ►

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5 Over time, 80% of the customers of a restaurant have given the restaurant an 'Excellent' rating.

The restaurant hires a new chef.

After one month, the new chef claims that the proportion of customers giving the restaurant an 'Excellent' rating has increased.

A random sample of 15 customers is taken and 14 customers give the restaurant an 'Excellent' rating.

5 (a) Test the new chef's claim, using a 10% level of significance.

[6 marks]

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5 (b) Describe, in the context of the test in part **(a)**, a Type II error.

[1 mark]

7

Turn over for the next question

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6 The marks scored in a maths test by a class of students are modelled by a normal distribution with mean μ and standard deviation σ.

5% of the students in the class scored more than 83 marks.

6 (a) Show that $83 - \mu = 1.6449\sigma$

[2 marks]

6 (b) 8% of the students in the class scored less than 10 marks.

Find the value of μ and the value of σ , giving your answers to three significant figures.

[5 marks]

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μ = _____ σ = _____

6 (c) The maximum possible mark for the test is 85 marks.

Explain why the normal distribution model used in parts **(a)** and **(b)** may not be a valid model.

[1 mark]

8

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7 (a) The diameters of pipes produced by machine *A* have a normal distribution with mean 14 millimetres and standard deviation 0.25 millimetres.

Following a power cut, a random sample of 25 pipes taken from machine *A* has a mean diameter of 13.892 millimetres.

Test whether the mean diameter of pipes produced by machine *A* has changed following the power cut, using a 2% level of significance.

Assume that the standard deviation is unchanged by the power cut.

[7 marks]

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7 (b) The diameters of pipes produced by machine *B* have a normal distribution.

A random sample of 9 pipes taken from machine *B* has a mean diameter of 14.02 millimetres and a standard deviation of 0.04 millimetres.

Test whether the mean diameter of pipes produced by machine *B* is higher than 14 millimetres, using a 10% level of significance.

[7 marks]

14

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8 The continuous random variable X has probability density function $\mathrm{f}(x)$ defined by

$$\mathrm{f}(x) = \begin{cases} k(x^2 - 7x + 6) & 1 \le x \le 6 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant.

8 (a) Show that $k = -\dfrac{6}{125}$

[3 marks]

8 (b) Show that $\mathrm{E}(X) = 3.5$

[3 marks]

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8 (c) Find $\mathrm{Var}(X)$

[5 marks]

Answer

11

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9 The heights of a particular type of plant have a normal distribution with mean 20 centimetres and variance 6.5536 cm^2

A random sample of 25 plants is taken.

9 (a) (i) Describe the distribution of the sample mean of the 25 plants.

[1 mark]

9 (a) (ii) Find the probability that the mean height of the plants is greater than 21 centimetres, giving your answer to three significant figures.

[3 marks]

Answer

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9 (b) (i) Describe the distribution of the total height of the sample of 25 plants.

[1 mark]

9 (b) (ii) Find the probability that the total height of the plants is greater than 525 centimetres, giving your answer to three significant figures.

[3 marks]

Answer

9 (c) Explain the link between your answers to parts **(a)(ii)** and **(b)(ii)**.

[2 marks]

10

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10 (a) A simple random sample is taken from car owners in a town.

10 (a) (i) State the population for this sample.

[1 mark]

10 (a) (ii) State a condition which must be satisfied by a simple random sample.

[1 mark]

10 (b) State the parameter used in calculating probabilities using the Poisson distribution.

[1 mark]

10 (c) State the parameters used in calculating probabilities using the normal distribution.

[2 marks]

5

END OF QUESTIONS

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